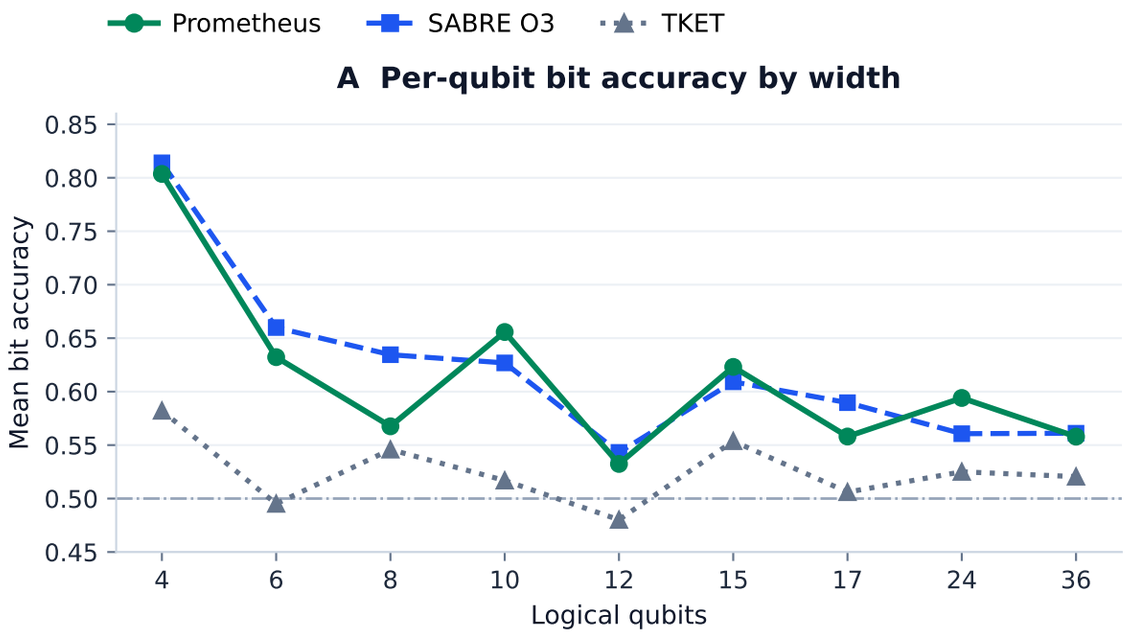
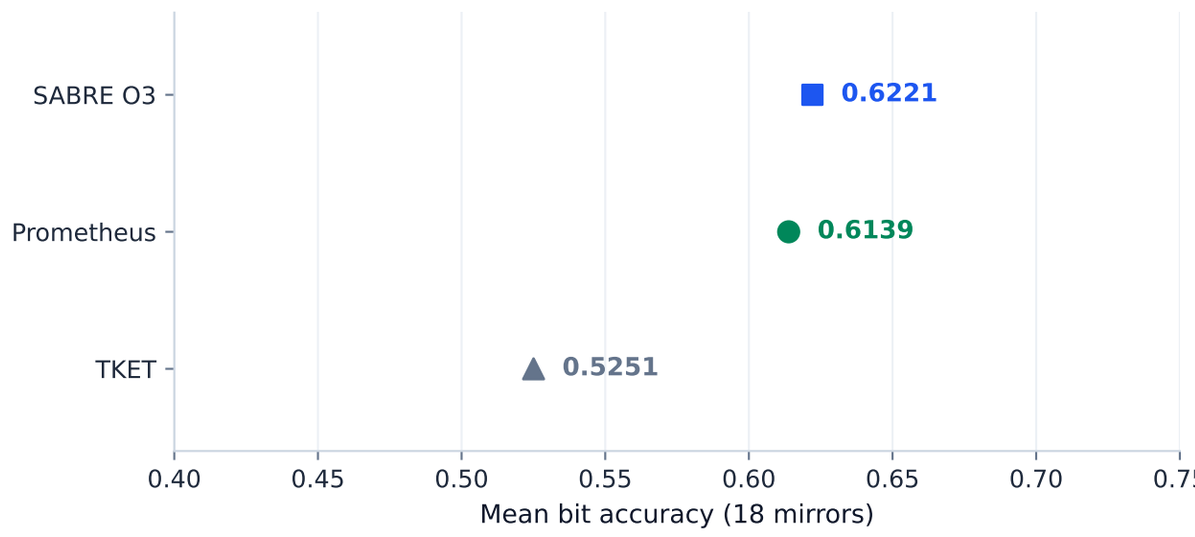
Prometheus: 4-36 qubit Clifford mirror validation

Rigetti Cepheus-1-108Q | 18 mirror instances (d=1, 2) + 12 SPAM controls | 66 circuits | 67,584 shots

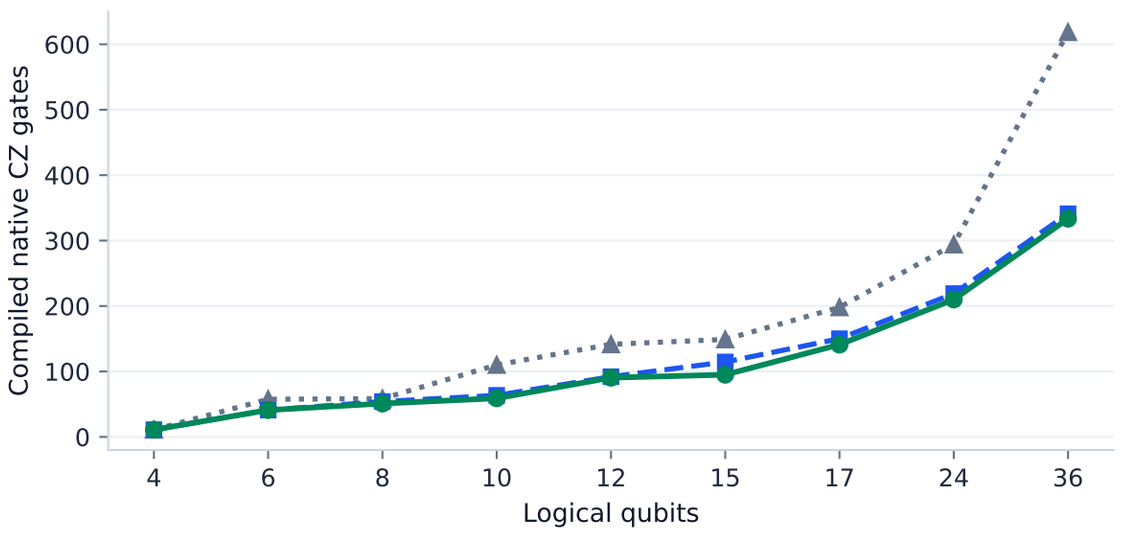
All 66 compiled physical circuits verified via exact StabilizerState simulation prior to execution.



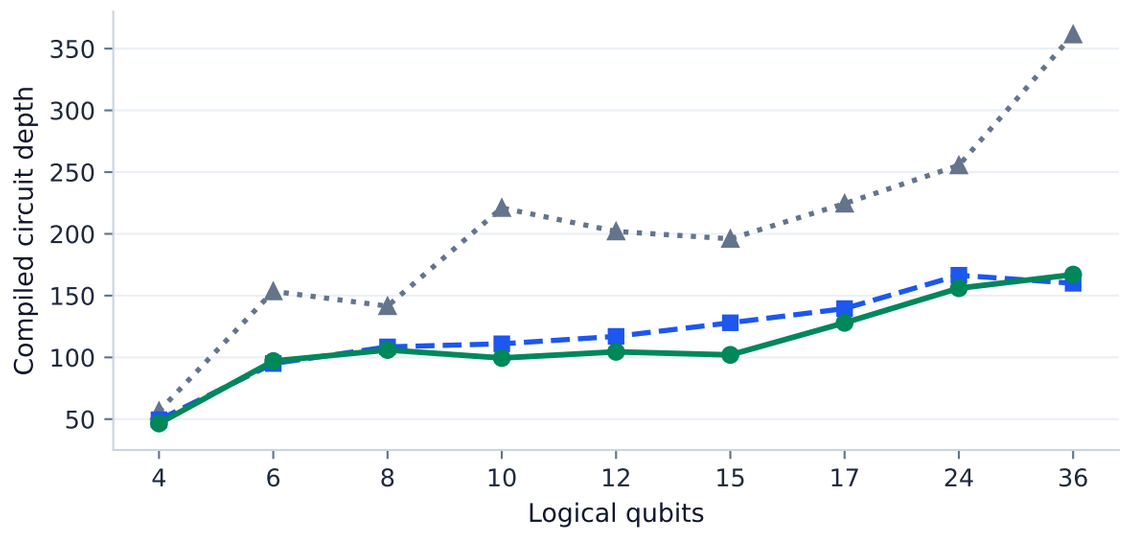
Averaged over d=1 and d=2 mirror passes per width; higher is better.



Overall mean across all 18 mirror instances (4-36 qubits).



Pooled mean: 114.67 Prometheus vs 120.67 SABRE (-4.97%) & 182.00 TKET (-37.00%).



Pooled mean: 111.83 Prometheus vs 119.44 SABRE (-6.37%) & 201.33 TKET (-44.45%).

Bit accuracy gain vs TKET: +16.91% relative (+0.0888 absolute, 16W-0T-2L)

Exact paired sign-flip tests, Holm-adjusted: p = 0.000366 vs TKET | p = 0.2737 vs SABRE (0.6139 vs 0.6221).

Prometheus achieves the lowest mean two-qubit gate count (114.67) and shallowest mean depth (111.83) across the 4-36Q sweep.